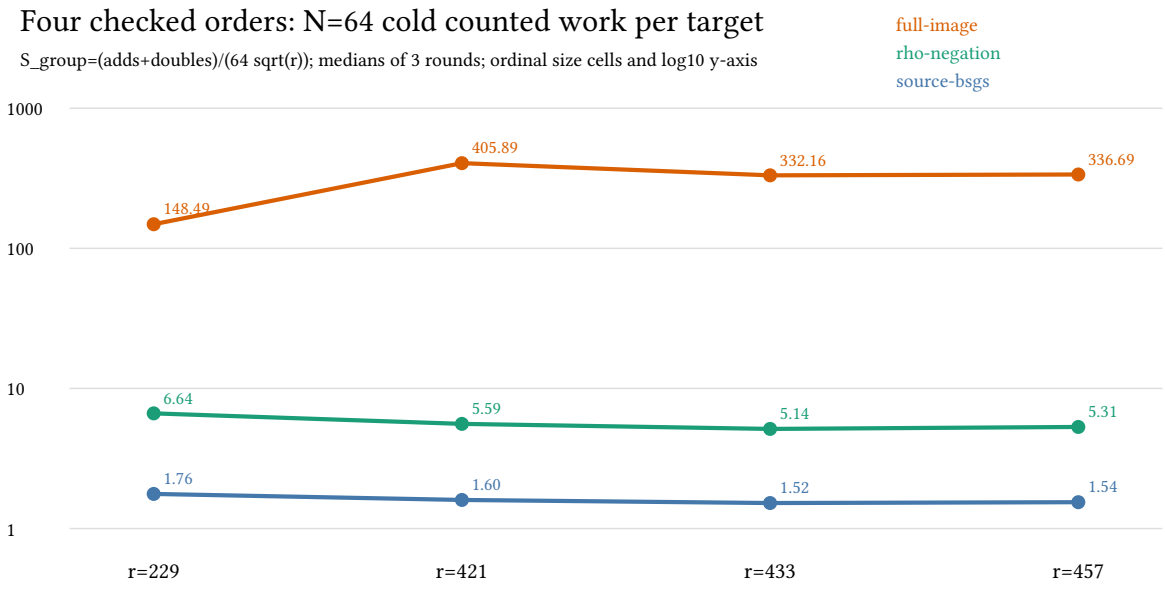
Four-order complete-solve panel

Status: 252/252 new cold children and 5,355/5,355 new scalar outputs passed independent Rust replay. Across the checked order-229 reference and three new prime-to-characteristic cells, 336 children and 7,140 scalar outputs are in the native count derivation. New witnesses: 37,061; all returned relation rows and scalar identities checked. The run spec and all source/binary/input digests were committed before measurement. Host: Darwin arm64, shared L0.

$S_{\text{group}} = \frac{\text{adds} + \text{doubles}}{N\sqrt{r}}$ is the cold counted group-operation total per target, with N=64 here. It includes common/setup, all queries and final verification. It is a **lower bound on complete portable S**: field work outside group calls and hash/table/row/alloc/process work have no pinned common conversion. The direct source negation-rho reference is on identical public targets; source BSGS is the memory control. The two-class generic collision floor is $\sqrt{\frac{\pi}{4}} \approx 0.8862$ per target.

Arm	$r = 229$	$r = 421$	$r = 433$	$r = 457$
Source projected	58.28	80.04	78.84	64.15
Isogeny-image	64.56	86.17	85.39	69.73
Destination-direct	69.56	81.83	68.04	72.28
Full rational image	148.49	405.89	332.16	336.69
Reusable source BSGS	1.76	1.60	1.52	1.54
Source negation rho	6.64	5.59	5.14	5.31
Rho A/A	6.64	5.59	5.14	5.31

Medians of three complete cold children per cell. Paired median full-image/rho group-count ratios are 22.355, 72.401, 64.275 and 63.389. In every cell, prefix and round, full-image exceeds BSGS and rho in each counted arithmetic axis: group work, Fp multiplications and Fp inversions. Native categories remain unpriced.



Source: 336 checked cold children, 7,140 scalar replays. Nonarithmetic work remains unpriced; L0 host timing is separate.

Relation geometry and remaining gates

The full rational-image table changes two-sum support but does not consistently increase it. The larger three cells have full-image support of 108, 106 and 112 keys versus destination-direct 112, 120 and 122. Each relation arm reached rank nine and recovered its requested scalar; the independent checker replayed 37,061 new relation witnesses. The full-image construction enumerates the entire rational source in every cold child, so its setup grows with the field. Its counted cost is 332–406 per target on the new cells, against rho's 5.14–5.59. The four orders span only a factor of two and use different cofactors; a stable exponent fit is not admitted.

r	full/direct keys	full relations/trials	direct relations/trials
229	98 / 90	2,329 / 5,389	2,628 / 6,776
421	108 / 112	2,457 / 9,475	2,348 / 8,853
433	106 / 120	2,393 / 9,595	2,234 / 8,113
457	112 / 122	2,376 / 9,556	2,339 / 8,830

The direct chi-coordinate resultant/root oracle is the next structural test: find useful destination columns without a full rational-source census, first prove exact decomposition coverage against independent enumeration, then price complete target solves. The pairing transfer has its own finite-field DLP and input-search cost gate. Complete portable pricing, L2-or-better interleaved timing with A/A and a second-host performance replay remain open. This run's CPU and wall meters are shared-host L0 descriptions; the canonical frontier and scoreboard graphs are unchanged.

The isolated-host access check found no configured isolab and a zero Standard On-Demand EC2 vCPU quota in five checked regions. A frozen Linux hosted-runner protocol is ready for independent-host replay; that virtual machine cannot admit L2 wall timing.

Source-linked records, run PDFs, exact hashes and accepted limits: `RESULTS_V2.md`, `BATCH_RUN_SPEC_V1.json`, and `TREND_V1.json`.